



Lecture 6: Fundamentals of AI agents

Business Applications (Betriebswirtschaftliche Anwendungen)

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Recap last week

Go to slido.com
and join with **# 1899**



Course content

1	Course Kickoff	7	AI agent ecosystem & Retro
2	Introduction to ERP systems	8	AI agents in business workflows
3	SAP ERP	9	Agentic ERP in S/4 HANA
4	Principles of ERP systems	10	OpenClaw & Test preparation
5	Modern ERP concepts	11	E-Test
6	Fundamentals of AI agents		

Instead: ERP system presentations

Learning objectives

At the end of the lecture, students are able to ...

- ➔ ... define **AI agents** and explain **core characteristics**
- ➔ ... distinguish different **types of AI agents**
- ➔ ... describe the **seven-layer architecture** and compare **popular frameworks**
- ➔ ... assess **typical challenges** of AI agent development for organizations





Agenda

1

Definition of AI Agents

2

Types of AI Agents

3

Technological Enablers

4

Architecture and Frameworks

5

Practical Challenges

Activity: What is an AI agent?

- Own brainstorming without researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)



AI agent definition (1/2)



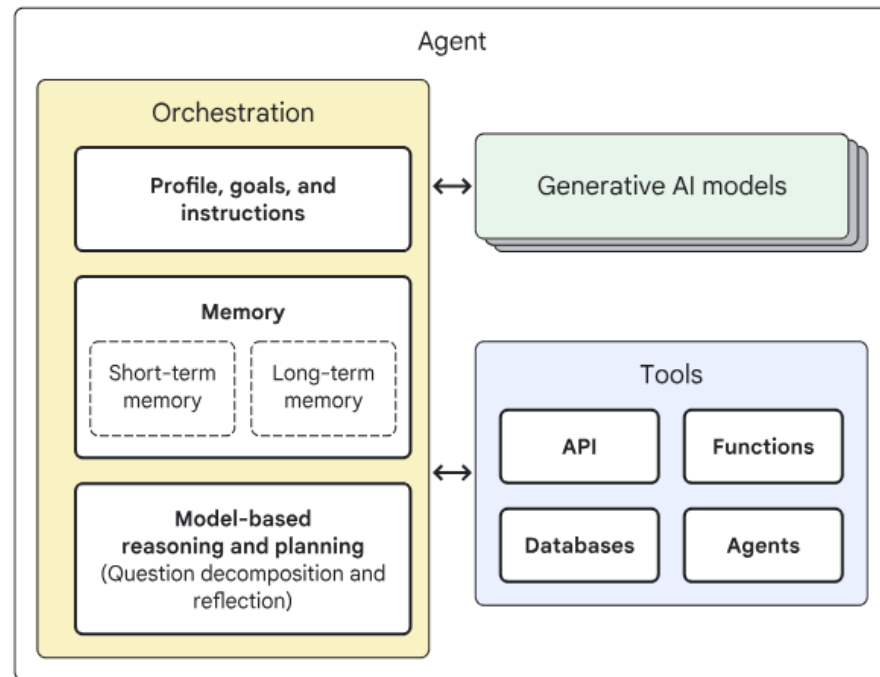
https://youtu.be/d0wUM8hlaxE?si=-hQgi_TCKLuMfoQC

AI agent definition (2/2)

“An *AI agent* is an application that achieves a goal by processing input, performing reasoning with available tools and taking actions based on its decisions. [...]

AI agents use function calling to format the input and ensure precise interactions with external tools.”

(Google Cloud)



Characteristics of AI agents (1/2)

1

Adaptability and Learning

Continuously refine knowledge through deep and transfer learning, generalize adaptability toward **human-like cognitive flexibility**

2

Multimodal Perception

Process **text, speech, images and sensory input** (e.g., radar or infrared signals), mirroring **multisensory biological input**

3

Reasoning and Problem-Solving

Logical deduction, causal inference and creative problem-solving using concepts from probabilistic reasoning

4

Social Intelligence and Collaboration

Understand **human emotions**, engage in **natural dialogue** and cooperate in **complex multi-agent systems**

5

Ethical Reasoning and Value Alignment

Reason about ethical and moral implications, ensure behaviors align with **human values and societal norms**

6

Meta-Learning and Self-Improvement

Improve **own learning** (“learning to learn”), rapidly **adapt to new tasks** without explicit human reprogramming

Characteristics of AI agents (2/2)

7

Explainability and Transparency

Provide **clear rationales for agent actions** based on built-in features, enabling **human auditing, trust-building and accountability**

8

Domain Agnosticism

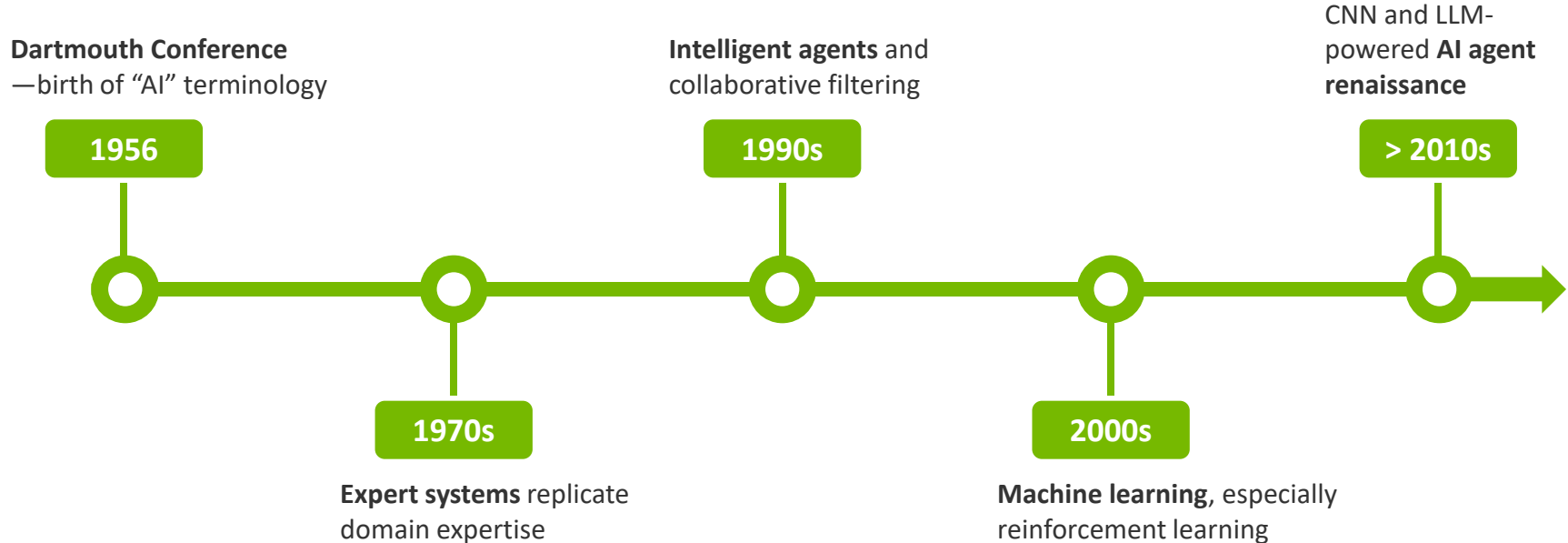
Transfer skills and knowledge across diverse fields (e.g., from healthcare to finance to creative tasks) to tackle various challenges

9

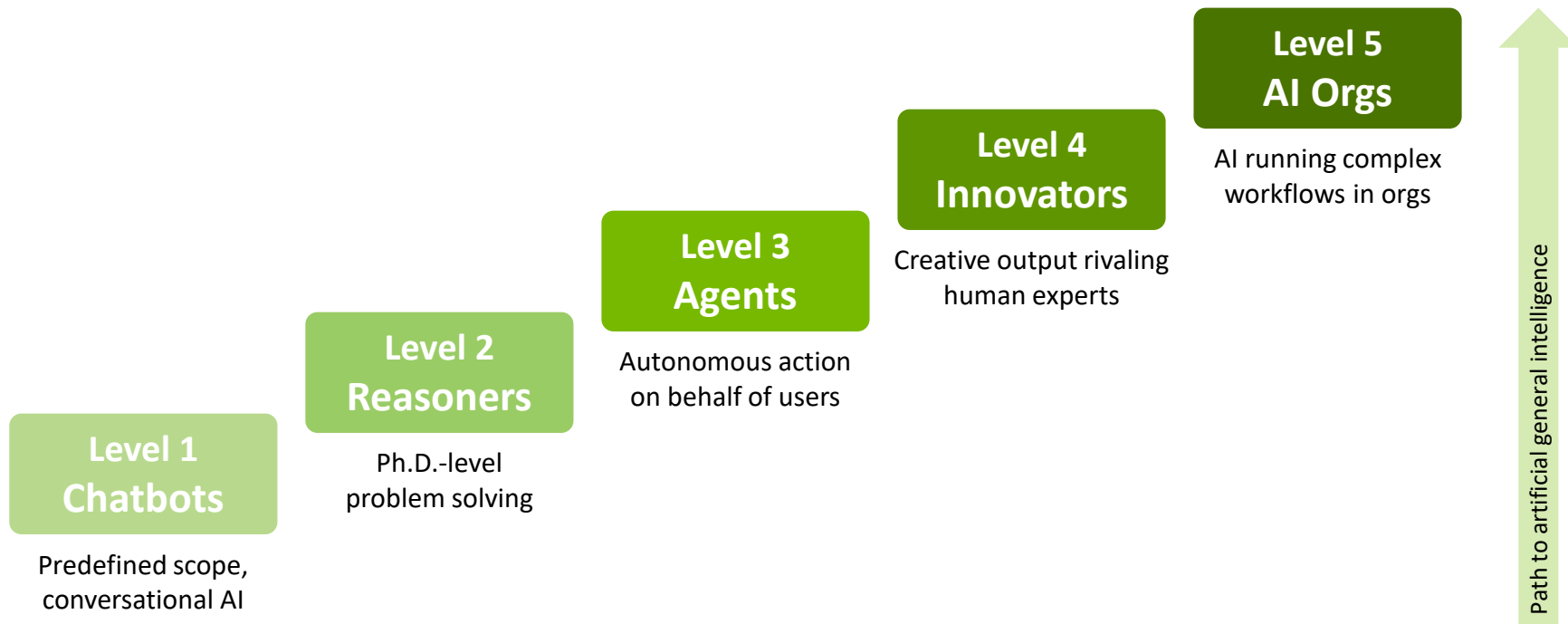
Embodied Intelligence

Physical embodiments of AI agents in robotics and IoT, interacting with the physical world (e.g., smart manufacturing, autonomous cars)

Historical trajectory of AI agents



OpenAI's classification of progress toward AGI





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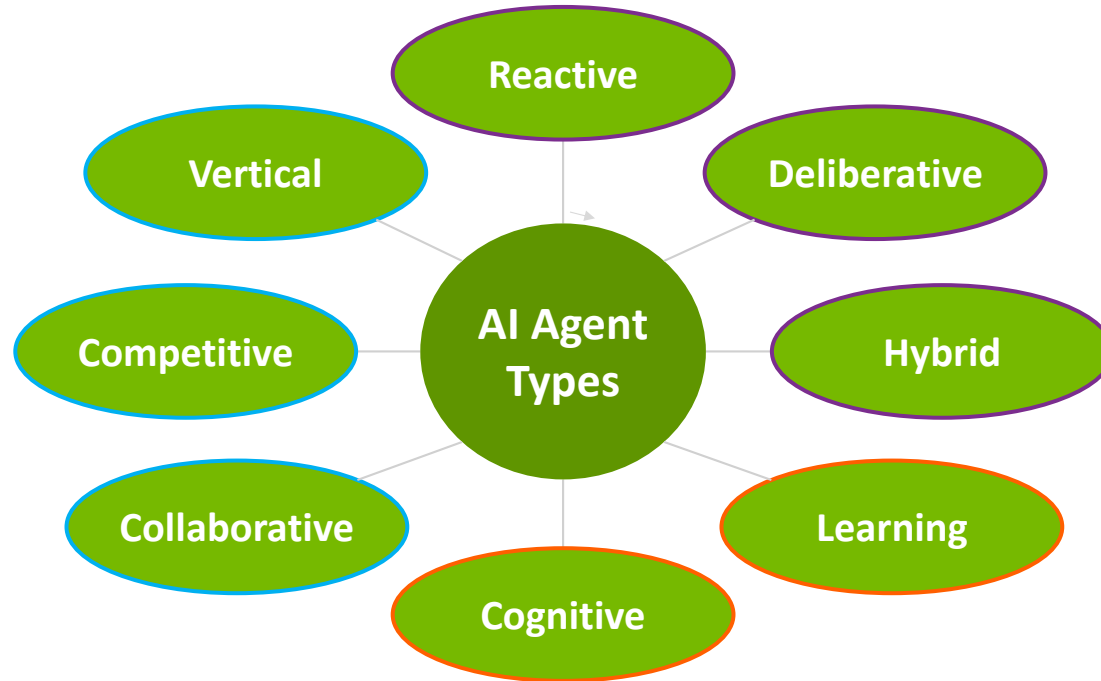
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Practical Challenges

Taxonomy of AI agents



Reactive vs. deliberative vs. hybrid agents

Reactive Agents

Respond directly to stimuli
based on predefined rules

- + Rapid response times due to minimal processing between input and action
- No learning or long-term planning

Use for: Frequently changing environments where quick decision-making is needed

Deliberative Agents

Reason and plan future actions
based on internal world models

- + Complex reasoning and problem-solving, long-term optimization
- High computational effort, "analysis paralysis"

Use for: Strategic planning, logistics, sophisticated games

- No such agent exists today -

Hybrid Agents

Combine reactive speed with deliberative planning depth

- + Ability to switch between reactive and deliberative modes based on situation
- Difficulty in balancing the two behavioral modes

Use for: Autonomous vehicles navigating urban environments, complex robotic systems

Learning vs. cognitive agents

Learning Agents

Improve own performance over time through experience and feedback by using machine learning techniques

- + Generalize from past experiences to handle novel situations
- Depend on training data (quality, quantity)

Use for: Recommendation systems, adaptive control systems, predictive maintenance

Cognitive Agents

Mimic human-like reasoning and problem-solving by using advanced NLP, knowledge representation and reasoning techniques

- + Ability for abstract reasoning and conceptual learning
- High computational effort, ethical concerns

Use for: Virtual assistants capable of complex dialogue, research and discovery systems

Collaborative vs. competitive vs. vertical agents

Collaborative Agents

Multiple agents working together in multi-agent systems

- + Capacity for distributed problem-solving and decision-making
- Coordination complexity, resource allocation

Use for: Settings where difficult tasks can be decomposed into subtasks for individual agents

Competitive Agents

Multiple agents competing in multi-agent systems

- + Each agent pursues own goal and anticipates other agents' behaviors
- Potential for escalating adversarial behaviors

Use for: Cybersecurity systems, trading in competitive markets

- "Adversarial agents" -

Vertical Agents

Agents with highly optimized performance for a specific task

- + Architecture optimized for single purpose (specialized algorithms/training data)
- Specialized hardware/real-time capabilities required

Use for: Specific applications (chess, medical diagnosis, etc.)

- "Domain-specific agents" -

Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes





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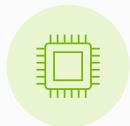
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Practical Challenges

Technological enablers of the AI agent renaissance



Computational Power

GPUs and TPUs enable high processing
e.g., NVIDIA hopper, Google TPU v4



NLP

Machines with almost human-like ability to understand and generate text
e.g., GPT-4



Big Data

Massive datasets fuel model training, advanced data analytics, IoT integration



Algorithmic Innovations

Breakthroughs in transformers, RL and self-supervised learning
e.g., AlphaZero



Interdisciplinary Insights

Neuromorphic computing and brain-inspired architectures
e.g., Intel Loihi 2 chip

Exemplary AI agent projects



OpenAI's Project “Operator” Agent

- Autonomous web navigation
- Deep research beyond simple retrieval
- Execution of multi-step actions
- Parallel operation of multiple agents

<https://youtu.be/gYqs-wUKZsM>



Stanford's Self-Taught Reasoner (STaR)

- Iterative self-improvement
- Generation of own training data
- Bootstrapping from only few examples
- Chain-of-Thought reasoning capabilities

<https://www.youtube.com/watch?v=rQLZgEgmQCc>

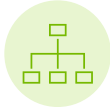
Some takeaways until here



Beyond simple chatbots: AI agents perceive, reason, act and learn autonomously



Evolution levels: From chatbots (level 1) to AI organizations (level 5)



Rich taxonomy: Various types of agents – from single agents to multi-agent systems



Confluence of technology: Hardware, data and algorithms to enable agentic AI



Societal transformation: AI agents as collaborators, not just tools – impacting societies



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Practical Challenges

Seven-layer AI agent architecture

L7 – Agent Ecosystem

Agent marketplace with pre-built agent-based business apps

L6 – Security and Compliance

Threat modeling, data security audits, compliance monitoring, etc.

L5 – Evaluation and Observability

Performance assessments, safety benchmarks (e.g., containment, alignment, robustness, interpretability), function execution tracing

L4 – Deployment and Infrastructure

Infrastructure-as-Code for automated deployment and configuration management, CI/CD for automated testing, load balancing

L3 – Agent Frameworks

Software packages for developing and managing AI agents (e.g., prompt management, CoT reasoning, RAG, debugging, error handling)

L2 – Data Operations

Vector DBs, data governance, ETL pipeline orchestration, etc.

L1 – Foundation Models

Core AI models (e.g., GPT-4, Claude, Gemini) with API endpoints

Popular AI agent frameworks

AutoGen

- Open-source framework from Microsoft
- Multi-agent conversations
- Flexible code execution
- Human-in-the-loop
- Docker-based sandboxing

LangGraph

- Open-source framework from LangChain Inc.
- Agent and multi-agent workflows
- Graph-based architecture
- StateGraph as central component
- Persistent state management

LlamaIndex

- Open-source data framework from Jerry Liu (run-lama)
- Package to connect data sources to LLMs
- Multi-source data connectors
- Document indexing
- Advanced RAG pipelines

AutoGPT

- Open-source AI application from Toran Bruce Richards
- Uses GPT-4 to create autonomous agents
- Minimal human intervention
- Memory mgmt., Internet access and code execution

Deep-Dive: AutoGen features

Feature	Description
Multi-agent conversations	AutoGen allows multiple GenAI Agents to converse and collaborate on solving complex tasks. Agents with different roles and capabilities can work together in dynamic systems. Example: A project manager agent, a code generator agent and a code reviewer agent collaborate.
Agent types and roles	• AssistantAgent: AI agent powered by an LLM, capable of processing requests, generating responses, answering questions and generating content. • UserProxyAgent: executes code and represents user input, allowing seamless integration of human feedback in conversations. These agents can be customized for various use cases
GroupChat functionality	AutoGen supports group conversations managed by a GroupChatManager, enabling multiple agents to interact and collaborate. The GroupChatManager ensures each agent contributes appropriately to the conversation, fostering diverse perspectives for complex problem-solving.
Code execution environments	AutoGen supports multiple environments for code execution: • Local execution for quick prototyping • Docker-based execution for added security and isolation • No-code execution focusing on language-based interactions

Deep-Dive: LangGraph features

Feature	Description
Graph-based architecture	LangGraph uses a graph-based architecture to represent agent workflows as a network of nodes and edges. • <i>Nodes</i> represent individual agents or processing steps. • <i>Edges</i> define transitions between nodes. • <i>State</i> maintains context and data flow through the graph, enabling complex workflows.
StateGraph and edge definitions	The central component of LangGraph is the StateGraph, initialized with a state schema. Edges define the flow between nodes, supporting conditional logic and complex branching for decision-making. Example: <pre>from langgraph.graph import StateGraph from langgraph.prebuilt import MessagesState graph = StateGraph (MessagesState)</pre>
Cycles and branching capabilities	LangGraph supports cycles in workflows, enabling iterative and looping processing and complex decision-making. This feature allows workflows to refine or explore multiple possibilities before reaching a conclusion, making it ideal for tasks requiring repeated refinement.

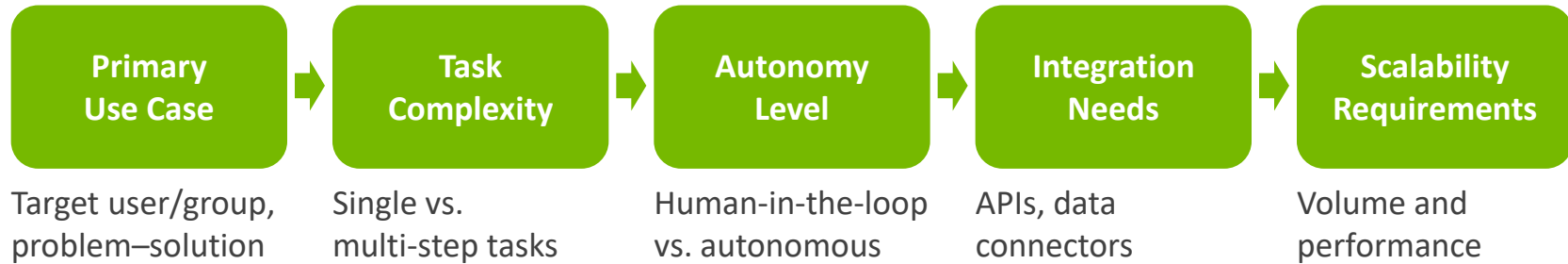
Deep-Dive: LlamaIndex features

Feature	Description
Data connectors and indexing	• LlamaIndex provides a wide range of connectors for integrating data sources, including files, APIs, and databases. • The indexing process utilizes advanced techniques, such as semantic indexing, for accurate and nuanced data retrieval beyond simple keyword matching.
Query and chat engines, event-based agent collaboration	• LlamaIndex offers query engines for natural language queries against indexed data. • These engines interpret complex questions, break them down into sub-queries, and retrieve relevant information. • Chat engines enable multi-turn conversations, ideal for building chatbots and interactive knowledge bases. • LlamaIndex uses an event-based approach for multi-agent collaborations.
Tool integration	LlamaIndex integrates with GenAI Agents, allowing them to use Python functions or LlamaIndex query engines as tools. For example, the QueryEngineTool allows agents to perform queries on specific data sources, expanding the system's capabilities.

Deep-Dive: AutoGPT features

Feature	Description
Autonomous operation	AutoGPT operates autonomously based on user-provided objectives. It plans and executes tasks independently, suitable for complex, multistep tasks that require minimal human intervention.
Memory management	AutoGPT features both long-term and short-term memory, enabling the agent to retain context over extended operations, learn from past actions and make informed decisions based on accumulated knowledge.
Internet access and file processing	AutoGPT can search the web for information and gather up-to-date data to support its tasks. It also supports file processing, including storing, summarizing, and analyzing files for document analysis and data synthesis.
Code execution	AutoGPT can write and execute code, making it suitable for programming tasks, software development, data analysis, and other computational problem-solving tasks

How to select an AI agent framework



Framework recommendations

AutoGen — Best for multi-agent systems with human oversight in enterprise applications

LangGraph — Best for complex, customized workflows integrating multiple tools

LlamaIndex — Best for data-intensive applications retrieving data from diverse sources

AutoGPT — Best for fully autonomous task execution with minimal human input



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Practical Challenges

Activity: What makes developing an AI agent difficult for orgs?

- Own brainstorming without researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)



Practical challenges in AI agent frameworks

Technical Complexity

Limited tech skills in orgs, outdated configs, no support

Software Integration

Incompatible data structures, limited API support

Scalability

Performance bottlenecks under high workloads

Security

Data breaches, malicious attacks, unauthorized access

Compliance

Regulatory standards like EU AI Act, UK AISI, HIPAA

Data Quality

Inconsistent data, siloed data, no data cleaning processes

Workforce

Shortage of talent in ML, software dev and data science

Costs

High costs for LLM API calls, infra, integration, labor

Some takeaways until here



Seven-layer architecture: A holistic blueprint from Foundation Models to Agent Ecosystem



Framework landscape: AutoGen, LangGraph, LlamaIndex, AutoGPT as popular examples



Framework choice: Task complexity and autonomy level as the first two criteria



Practical challenges: Technical complexity, scalability, data quality, costs and more



Security-first mindset: Robust security and compliance should be a key priority

Thank you for your attention



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